



AI at the Sella Turcica: Multi-Model Large Language Model Evaluation in Pituitary Adenomas

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ARTICLE INFO

Keywords:

Large language model
Pituitary adenoma
Neuroendocrinology
ChatGPT-5
Claude opus
Gemini

ABSTRACT

Introduction: Large language models (LLMs) are explored as clinical decision-support tools in complex medical fields. However, their reliability and clinical usefulness in multidisciplinary management of pituitary adenomas remain insufficiently evaluated using validated, clinician-based frameworks.

Research question: Do LLMs differ in informational quality, clinical reasoning, and expert satisfaction when applied to pituitary adenoma-related clinical scenarios?

Materials and methods: A prospective comparative study evaluated three LLMs: ChatGPT-5.0, Claude Opus 4.1, and Gemini 2.5 Flash. A standardized prompt set covering general knowledge, surgical decision-making, endocrine evaluation, patient education, and MRI-based scenarios was submitted to each model identically. Outputs were anonymized and independently assessed by 10 board-certified doctors using three validated instruments: the Quality Assessment of Medical Artificial Intelligence (QAMAI), the Artificial Intelligence Performance Instrument (AIPI), and the Artificial Intelligence Satisfaction and Performance Evaluation Questionnaire (AISPE-Q).

Results: Claude Opus 4.1 achieved the highest performance across all major domains. Aggregate QAMAI scores were highest for Claude Opus 4.1 (4.39 ± 0.66), compared with ChatGPT-5.0 (4.12 ± 0.74) and Gemini 2.5 Flash (4.07 ± 0.76 ; $p = 0.018$). Clinical reasoning assessed by AIPI was superior for Claude Opus 4.1 versus Gemini 2.5 Flash and ChatGPT-5.0. Strong correlations were observed between informational quality, reasoning performance, and satisfaction.

Discussion and conclusion: LLMs exhibit significant variability in performance when managing pituitary adenomas. Claude Opus 4.1 demonstrated the highest levels of informational quality, reasoning depth, and expert trust. While LLMs may serve as supportive adjuncts in multidisciplinary pituitary care, structured evaluation and expert oversight remain essential before clinical integration.

Level of evidence: 2 – Prospective comparative diagnostic accuracy study.

1. Introduction

Artificial intelligence (AI) has advanced rapidly in healthcare applications, particularly for diagnostic support and clinical decision-making (Obuchowicz et al., 2025; Khosravi et al., 2024). Large language models (LLMs) and machine learning frameworks have demonstrated value in synthesizing complex clinical and radiological data,

enhancing pattern recognition, and supporting evidence-based recommendations across various specialties (Maity and Saikia, 2025; Lin et al., 2025; Merchant et al., 2025). Despite this progress, the integration of AI into the assessment and management of multidisciplinary skull-base lesions and neurosurgical care remains limited (Patil et al., 2024; Sales et al., 2025a).

Pituitary adenomas (PA) are the most common sellar neoplasms,

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<https://doi.org/10.1016/j.bas.2026.105997>

Received 27 December 2025; Received in revised form 4 February 2026; Accepted 28 February 2026

Available online 6 March 2026

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comprising approximately 10–15% of intracranial tumors. They cause morbidity through mass effect, including visual impairment and cranial neuropathies, or through hormonal hypersecretion or deficiency (Jesser et al., 2014; Banskota and Adamson, 2021; Karimian-Jazi, 2019). Their management is inherently multidisciplinary, requiring integration of clinical findings, hormonal evaluation, neuro-ophthalmologic assessment, and neuroimaging. Despite established guidelines, variability in practice persists due to the involvement of multiple specialties. LLMs and AI-based tools may help bridge interdisciplinary decision-making and support patient-centered care by complementing existing management pathways (Aluyi-Osa et al., 2025; Zheng et al., 2025; Dai et al., 2021).

In this context, LLMs hold potential as adjunctive tools for both clinicians and patients (Aliyeva et al., 2025). Their capacity to analyze complex clinical inputs—ranging from hormone profiles to MRI interpretations—and to communicate results in both professional and layperson-friendly language, makes them attractive for deployment in pituitary adenoma management. However, few prior studies have systematically evaluated the performance of advanced LLMs in this domain (Ba et al., 2022; Akinci D'Antonoli et al., 2024). Reliable and accurate LLMs have the potential to improve interdisciplinary patient care in resource-constrained settings, support greater adherence to evidence-based guidelines, and strengthen patient engagement through clear educational messaging.

The present study provides a comparative evaluation of three state-of-the-art LLMs—ChatGPT-5.0 (OpenAI) (OpenAI, 2025), Claude Opus (Anthropic (2025)), and Gemini 2.5 Flash (Google (2025))—in the diagnosis and management of pituitary adenomas. Responses were assessed using three validated evaluation instruments (Vaira et al., 2024a; Lechien et al., 2024; Aliyeva et al., 2026; Neuropathology).

The primary hypothesis of this study is that significant differences exist among LLMs in the diagnostic accuracy, clarity, and clinical applicability of their responses. The secondary hypothesis is that higher objective LLMs performance will positively correlate with user satisfaction. By systematically comparing multiple LLMs across diverse pituitary adenoma scenarios, this work aims to provide critical insights into the future role of AI in multidisciplinary PA management.

2. Materials and Methods

2.1. Study design

This prospective, comparative study was designed to investigate the clinical performance and perceived utility of three advanced LLMs (OpenAI, 2025; Anthropic, 2025; Google, 2025) in the diagnosis and management of pituitary adenomas. A structured question pool was developed, comprising 20 general questions, physician-directed queries, five case-based clinical scenarios, and five representative MRI cases sourced from “*Radiopaedia*” (Neuropathology; O'Donnell et al.; Gaillard; Tsoriev; Rasuli) (Supplementary Material 1). For MR findings, one representative sagittal and one representative coronal view have been used. This question-and-case-scenario pool was curated by a multidisciplinary team of otolaryngologists, endocrinologists, neurologists, ophthalmologists, and neurosurgeons to ensure clinical relevance and alignment with guidelines.

Data Handling and Ethical Considerations. No real patient data or personally identifiable information were utilized in this study. All clinical scenarios, hormonal panel interpretations, and radiological findings were generated as hypothetical yet clinically realistic cases designed to mirror typical presentations of pituitary adenomas. To approximate real-world multidisciplinary workflows, scenarios were designed to reflect typical decision points in pituitary adenoma care (symptom-based presentation, endocrine workup, surgical planning, patient counseling, and MRI-based decision framing) and were curated by a multidisciplinary team to align with guideline-relevant reasoning. This approach aimed to standardize inputs across models while preserving

clinically representative complexity across specialties.

To systematically evaluate the responses, three validated instruments were used: the *Quality Assessment of Medical Artificial Intelligence (QAMAI)*, the *Artificial Intelligence Performance Instrument (AIPI)*, and the *Artificial Intelligence Satisfaction and Performance Evaluation Questionnaire (AISPE-Q)* (Vaira et al., 2024a; Lechien et al., 2024; Aliyeva et al., 2026).

QAMAI was used to objectively assess the accuracy, clarity, relevance, completeness, usefulness, and overall informational quality of AI-generated responses, providing a standardized framework for comparing content reliability across models (detailed scoring methodology in Supplementary Material 2).

AIPI was applied to evaluate higher-order clinical reasoning, including diagnostic coherence and management logic, to discriminate among models in complex, multidisciplinary pituitary adenoma decision-making scenarios (instrument structure and scoring criteria are described in Supplementary Material 3).

AISPE-Q was employed to capture expert-perceived satisfaction and clinical usability of AI outputs, allowing correlation of subjective trust with objective performance metrics (full questionnaire and evaluation process provided in Supplementary Material 4)

Together, these tools enabled a multidimensional assessment of accuracy, clinical reasoning, completeness, and user satisfaction.

LLMs Model Versions and Access Conditions. All three LLMs were accessed through their official platforms under default configurations during September–October 2025. Because the models were accessed via their official user platforms, generation parameters (e.g., temperature/top-p) and random seed control were not user-adjustable or reportable in this study. Accordingly, outputs may exhibit run-to-run variability even when identical prompts are used. All prompts were executed within this fixed evaluation window to provide a time-stamped snapshot of model behavior under contemporaneous deployments.

These included ChatGPT-5.0, Claude Opus 4.1, and Gemini 2.5 Flash (OpenAI, 2025; Anthropic, 2025; Google, 2025). No fine-tuning, plug-ins, or memory-based personalization was applied to maintain standardization, reproducibility, and fairness across all models.

Blinding and Evaluation Protocol. All responses generated by the LLM systems were anonymized and randomized before being distributed to evaluators. A panel of 10 board-certified neurosurgeons and otolaryngologists, each with expertise in skull base surgery and pituitary disorders, independently assessed the outputs. Evaluators were blinded to the LLMs' identities (as X, Y, Z) to minimize bias. The outputs generated by the three LLMs were independently assessed (Supplementary Material 5 and 6)

2.2. Data collection

A total of **20 prompts** were administered: **3** general knowledge questions (A), **3** surgical decision-making questions (B), **4** patient-education questions (C), **5** clinical scenarios (neurological, radiological, ophthalmological, endocrinological, and a hormonal-panel case) (D), and **5** MRI-based items (E). Each prompt was entered identically into all platforms using a standardized protocol. The full standardized prompt set (verbatim wording and formatting for each item): I am conducting research and will evaluate the answers from large language models (AI at the Sella Turcica: Multi-Model Large Language Model Evaluation in Pituitary Adenomas). I would like you to write an answer of no more than three sentences and 500 words. Is that ok?) is provided in Supplementary Material 1, with explicit documentation of the identical prompt header and constraints applied across all models. No system-level or hidden instructions were used; all models received the same user-level prompt text, which is archived and versioned in the Supplementary Material for independent replication. To simulate realistic interactions, responses were limited to 3 sentences and 500 words. Every question started as a new prompt, not as a continuous prompt. All inputs and outputs were de-identified before analysis.

For each MRI-based prompt, the same standardized question was applied across models, using two de-identified images (one sagittal and one coronal) from publicly available teaching cases (Radiopaedia), and models were asked to (Obuchowicz et al., 2025) infer the most likely diagnosis from imaging appearance and (Khosravi et al., 2024) propose an appropriate management plan within the three-sentence/500-word constraint. These MRI-based items, therefore, assess text-guided interpretation and clinical reasoning from representative images, not autonomous radiological assessment.

2.3. Statistical analysis

Descriptive statistics (mean, standard deviation, minimum, and maximum values) were calculated for all evaluation domains across the three LLM models. Between-model differences were analyzed using repeated-measures analysis of variance, with Bonferroni-adjusted pairwise t-tests applied for post hoc comparisons when significant effects were observed. Effect sizes were calculated using Cohen's d to quantify the magnitude of between-model differences across core domains, including Accuracy, Clarity, Completeness, and Usefulness. Inter-rater reliability was assessed using Fleiss' kappa for categorical agreement and intraclass correlation coefficients (ICC) for continuous ratings. Associations between objective performance metrics (QAMAI and AIPI) and evaluator satisfaction (AISPE-Q) were examined using Pearson correlation coefficients. All statistical analyses were performed using IBM SPSS Statistics (version 29.0). A two-tailed p-value <0.05 was considered statistically significant.

3. Results

ChatGPT-5, Claude Opus 4.1, and Gemini 2.5 Flash were evaluated using a structured expert-based framework that incorporated QAMAI domain scoring, AIPI clinical reasoning assessment, and AISPE-Q evaluator satisfaction metrics (See Table 1).

3.1. Domain-level QAMAI performance

Across the six QAMAI domains, Claude Opus 4.1 demonstrated the highest mean scores in all categories. *Accuracy of Information* showed a statistically significant model effect, with Claude Opus 4.1 scoring 4.41 ± 0.70 , compared with 4.03 ± 0.92 for ChatGPT-5 and 3.95 ± 0.85 for Gemini 2.5 Flash ($p = 0.009$).

Quality scores were numerically higher for Claude Opus 4.1 at 4.22 ± 0.62 , followed by Gemini 2.5 Flash at 4.19 ± 0.67 and ChatGPT-5 at 4.10 ± 0.76 , although this difference did not reach statistical significance.

Clarity demonstrated high overall performance across models, with Claude Opus 4.1 achieving 4.56 ± 0.55 , compared with 4.46 ± 0.60 for ChatGPT-5 and 4.38 ± 0.63 for Gemini 2.5 Flash. Between-model differences were not statistically significant.

Relevance scores revealed a significant model effect, with Claude

Opus 4.1 scoring 4.34 ± 0.77 , compared with 4.16 ± 0.67 for ChatGPT-5 and 3.95 ± 0.76 for Gemini 2.5 Flash ($p = 0.014$).

Usefulness followed a similar pattern, with Claude Opus 4.1 achieving 4.39 ± 0.64 , exceeding ChatGPT-5 at 4.12 ± 0.74 and Gemini 2.5 Flash at 4.08 ± 0.76 ($p = 0.032$).

Completeness showed the largest separation among models. Claude Opus 4.1 scored 4.41 ± 0.64 , while ChatGPT-5 and Gemini 2.5 Flash scored 3.85 ± 0.77 and 3.86 ± 0.87 , respectively ($p = 0.012$).

Fig. 1a illustrates the relative distribution of mean QAMAI domain scores using a radar plot, highlighting the consistent dominance of Claude Opus 4.1 across all informational dimensions. Fig. 1b shows the same domain scores with standard deviation error bars, demonstrating lower variability for Claude Opus 4.1 than for the other models.

3.2. Aggregate QAMAI performance

When QAMAI domains were aggregated into a composite score, Claude Opus 4.1 achieved the highest overall QAMAI mean at 4.39 ± 0.66 , compared with 4.12 ± 0.74 for ChatGPT-5 and 4.07 ± 0.76 for Gemini 2.5 Flash. Repeated-measures ANOVA confirmed a significant overall model effect on aggregate QAMAI performance ($F = 4.12$, $p = 0.018$), with a moderate effect size ($\eta^2 = 0.14$). Pairwise comparisons demonstrated a statistically significant advantage of Claude Opus 4.1 over ChatGPT-5 ($p = 0.006$, Cohen's $d = 0.67$) and over Gemini 2.5 Flash ($p = 0.010$, Cohen's $d = 0.61$). No statistically significant difference was observed between ChatGPT-5 and Gemini 2.5 Flash.

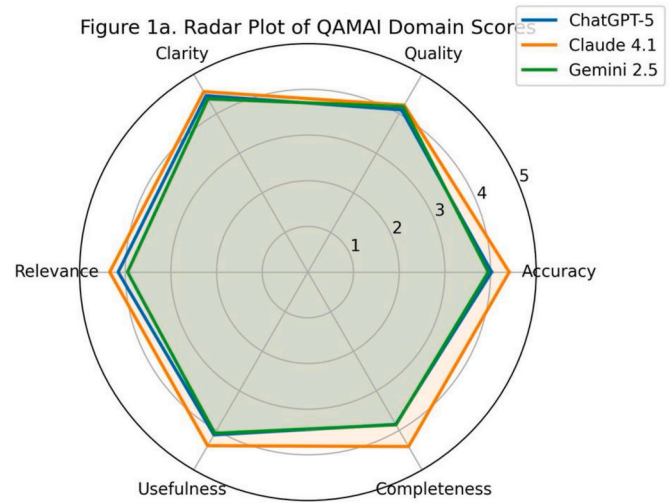


Fig. 1A. Radar plot of QAMAI domain scores.

Table 1

Integrated Evaluation of Large Language Models

Data are presented as Mean ± SD across 10 independent evaluators.

Evaluation Parameter		ChatGPT-5 (OpenAI)	Claude Opus 4.1 (Anthropic)	Gemini 2.5 Flash (Google)	p-value (3-LLM comparison)	Best-Performing LLM
QAMAI (Supplementary 1)	Accuracy of Information	4.03 ± 0.92	4.41 ± 0.70	3.95 ± 0.85	0.009*	Claude Opus 4.1
	Quality	4.10 ± 0.76	4.22 ± 0.62	4.19 ± 0.67	0.527	-
	Clarity	4.46 ± 0.60	4.56 ± 0.55	4.38 ± 0.63	0.462	-
	Relevance	4.16 ± 0.67	4.34 ± 0.77	3.95 ± 0.76	0.014*	Claude Opus 4.1
	Usefulness	4.12 ± 0.74	4.39 ± 0.64	4.08 ± 0.76	0.032*	Claude Opus 4.1
	Completeness	3.85 ± 0.77	4.41 ± 0.64	3.86 ± 0.87	0.012*	Claude Opus 4.1
AIPI Total (0–20) (Supplementary 2)		14.9 ± 1.97	16.9 ± 2.18	16.4 ± 1.17	0.038*	Claude Opus 4.1
AISPE-Q Total (0–10) (Supplementary 3)		8.3 ± 1.06	8.7 ± 0.95	7.9 ± 0.74	0.044*	Claude Opus 4.1

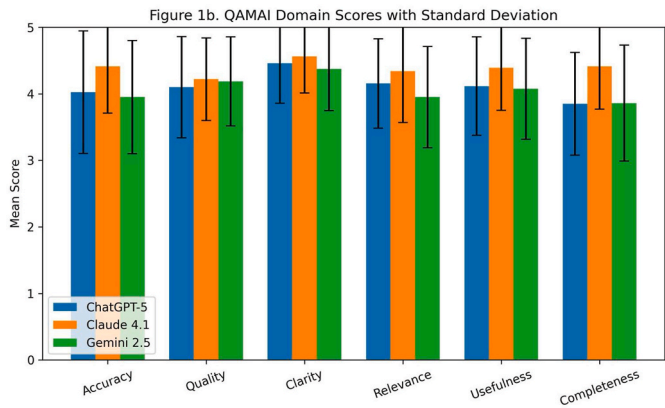


Fig. 1B. QAMAI domain scores with error bars.

- Claude Opus 4.1 has **higher means with relatively tight SDs**, suggesting both stronger performance and evaluator agreement.
 - Gemini shows wider SDs in several domains, reflecting greater response variability.
- Key visual takeaway

3.3. Clinical reasoning assessment using AIPI

Clinical reasoning performance, as evaluated by the AIPI instrument, differed significantly across models. Claude Opus 4.1 achieved the highest total AIPI score at 16.9 ± 2.18 , followed by Gemini 2.5 Flash at 16.4 ± 1.17 and ChatGPT-5 at 14.9 ± 1.97 ($p = 0.038$). Fig. 2 illustrates the comparative distribution of AIPI and AISPE-Q scores across models, demonstrating the consistent superiority of Claude Opus 4.1 in both objective reasoning performance and evaluator-perceived quality.

3.4. Evaluator satisfaction assessed by AISPE-Q

AISPE-Q scores revealed statistically significant differences among models. Claude Opus 4.1 achieved the highest evaluator satisfaction score at 8.7 ± 0.95 , compared with 8.3 ± 1.06 for ChatGPT-5 and 7.9 ± 0.74 for Gemini 2.5 Flash ($p = 0.044$). The close alignment between AISPE-Q satisfaction scores and both QAMAI and AIPI outcomes supports the internal validity of the evaluation framework.

3.5. Correlation analysis between evaluation instruments

Strong positive correlations were observed between aggregate QAMAI and AIPI scores ($r = 0.71$, $p < 0.001$), indicating that higher

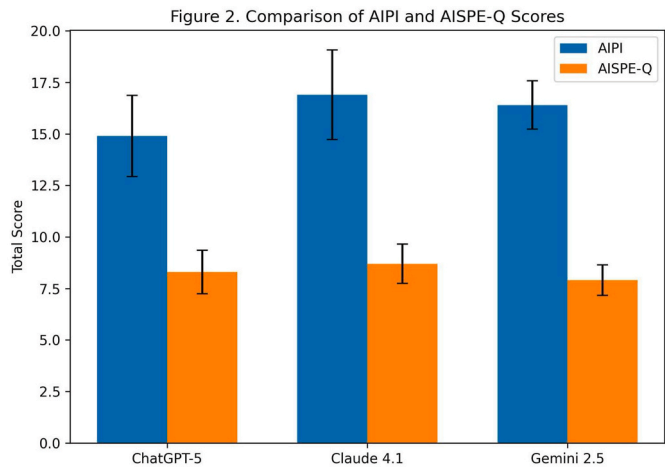


Fig. 2. Comparison of AIPI and AISPE-Q scores across LLMs.

informational quality was closely associated with stronger clinical reasoning performance. QAMAI scores also demonstrated a moderate-to-strong correlation with AISPE-Q satisfaction scores ($r = 0.64$, $p = 0.002$). AIPI and AISPE-Q were moderately correlated ($r = 0.59$, $p = 0.004$). Fig. 3 presents a comparison matrix that visualizes the relationships among QAMAI, AIPI, and AISPE-Q across models, highlighting consistent positive associations among informational quality, clinical reasoning, and evaluator satisfaction. The matrix uses a color gradient to represent the strength and direction of correlations between evaluation instruments. Warmer colors indicate stronger positive correlations, while cooler colors indicate weaker associations (See Table 2).

3.6. Inter-rater reliability

Inter-rater reliability analysis demonstrated good to excellent agreement across evaluators. ChatGPT-5 achieved good reliability with an ICC(2,k) of 0.81 ($p < 0.001$). Claude Opus 4.1 demonstrated excellent reliability with an ICC of 0.86 ($p < 0.001$). Gemini 2.5 Flash showed good reliability with an ICC of 0.78 ($p < 0.001$). These findings confirm that observed between-model differences are unlikely to be driven by evaluator inconsistency.

4. Discussion

Reliable, reproducible, and interpretable LLM-generated medical information is increasingly critical in complex, high-risk clinical settings that demand coordinated multidisciplinary decision-making (Savage et al., 2025; Fahim et al., 2025). PA management was evaluated as a high-complexity model, revealing strong correlations between informational quality, clinical reasoning, and expert satisfaction ($r = 0.71$ – 0.64), with Claude Opus 4.1 achieving the highest overall performance (AIPI 16.9 ± 2.18 ; AISPE-Q 8.7 ± 0.95). These findings frame the study as an assessment of alignment between informational quality, reasoning robustness, and perceived clinical trustworthiness in LLM-assisted pituitary care. Also, our results should be interpreted as a benchmark for three LLM deployments available in September–October 2025, rather than as a fixed estimate of model capability. Periodic re-evaluation using the same standardized prompt set and scoring instruments is warranted to monitor time-dependent performance drift and maintain clinical relevance.

The QAMAI tool was initially validated by Vaira et al. as a unidimensional instrument assessing informational quality across six domains, demonstrating high internal consistency (Cronbach's $\alpha = 0.837$),

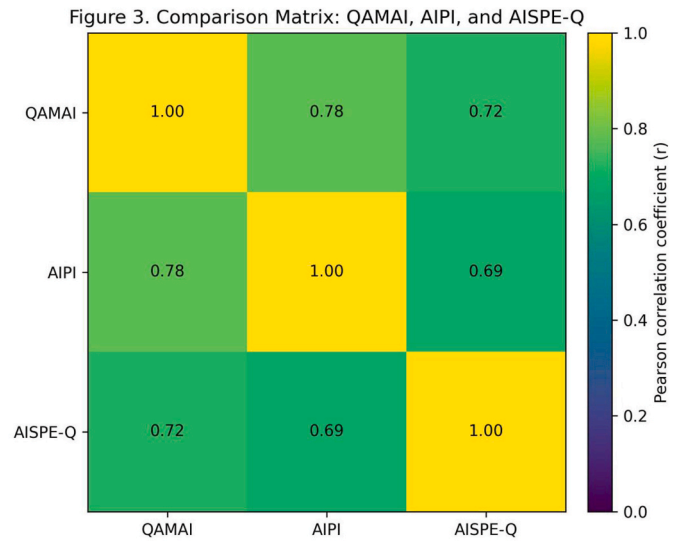


Fig. 3. Comparison matrix: QAMAI, AIPI, and AISPE-Q

Table 2
Advanced statistical summary (effect sizes, correlations, and inter-rater reliability).

Comparison/ Metric	Statistical Test	F/t/r	p-value	Effect Size	Interpretation
Overall model effect across QAMAI domains	Repeated- measures ANOVA	F = 4.12	0.018	$\eta^2 =$ 0.14	Moderate overall model effect
ChatGPT-5 vs Claude Opus 4.1 (QAMAI total)	Paired <i>t</i> - test	<i>t</i> = −3.01	0.006	Cohen's <i>d</i> = 0.67	Moderate–large advantage for Claude
ChatGPT-5 vs Gemini 2.5 Flash (QAMAI total)	Paired <i>t</i> - test	<i>t</i> = 1.42	0.17	Cohen's <i>d</i> = 0.32	Small, non- significant difference
Claude Opus 4.1 vs Gemini 2.5 Flash (QAMAI total)	Paired <i>t</i> - test	<i>t</i> = 2.74	0.010	Cohen's <i>d</i> = 0.61	Moderate advantage for Claude
QAMAI ↔ AIPI	Pearson correlation	<i>r</i> = 0.71	<0.001	$r^2 =$ 0.50	Strong correlation
QAMAI ↔ AISPE-Q	Pearson correlation	<i>r</i> = 0.64	0.002	$r^2 =$ 0.41	Moderate–strong correlation
AIPI ↔ AISPE-Q	Pearson correlation	<i>r</i> = 0.59	0.004	$r^2 =$ 0.35	Moderate correlation
Inter-rater reliability – ChatGPT-5	ICC(2,k)	ICC = 0.81	<0.001	—	Good reliability
Inter-rater reliability – Claude Opus 4.1	ICC(2,k)	ICC = 0.86	<0.001	—	Excellent reliability
Inter-rater reliability – Gemini 2.5 Flash	ICC(2,k)	ICC = 0.78	<0.001	—	Good reliability

excellent inter-rater reliability (ICC = 0.983), and strong test–retest stability (*r* = 0.876), confirming its robustness as an objective quality metric for AI-generated medical information (Vaira et al., 2024a). While Vaira et al. validated QAMAI as a stable measure of informational quality, our study extends its clinical relevance by demonstrating that higher aggregate QAMAI scores differentiate model performance and correlate with clinical reasoning (AIPI) and expert satisfaction (AISPE-Q) in PA management.

Vaira et al. reported high diagnostic accuracy of AI-generated content (87.2%) but markedly lower completeness (56.7%), with aggregate QAMAI scores ranging from 26.24 ± 1.26 to 28.24 ± 0.77 across platforms (*p* < 0.001). In contrast, our PA study demonstrated clearer model stratification within a single high-complexity domain, with Claude Opus 4.1 achieving higher accuracy (4.41 ± 0.70), completeness, and aggregate QAMAI performance. Unlike prior QAMAI applications, informational quality in our study showed strong correlations with clinical reasoning (AIPI *r* = 0.71) and evaluator satisfaction (AISPE-Q *r* = 0.64), highlighting the added value of integrated reasoning assessment in multidisciplinary pituitary care.

Claude Opus 4.1 demonstrated consistent superiority across all informational domains, with higher mean scores, lower variability, and a higher aggregate QAMAI score (4.39 ± 0.66; *d* = 0.61–0.67), indicating greater informational stability for longitudinal pituitary care, in line with evidence that accuracy and consistency are critical for safe AI-supported clinical decision-making as emphasized by Rosen et al. and Alharbi et al. (Rosen et al., 2025; Alharbi et al., 2025).

Lechien et al. validated the AIPI as a reliable clinical reasoning metric, demonstrating acceptable internal consistency (Cronbach's α = 0.754), moderate test–retest reliability (*rs* = 0.486, *p* = 0.001), and significant convergent validity with the modified OCAT (*rs* = 0.319, *p* = 0.044) when applied to ChatGPT in clinical cases (Lechien et al., 2024; Vaira et al., 2024b). In our pituitary adenoma study, higher AIPI scores were associated with superior information quality in a high-complexity neurosurgical–endocrine context, supporting AIPI's clinical relevance for multidisciplinary, high-risk decision-making.

In the subsequent multi-model evaluation by Pourhoseingholi et al., AIPI successfully differentiated reasoning performance among LLMs, including ChatGPT, Bing, and Gemini, confirming its ability to detect clinically relevant variability across AI systems (Pourhoseingholi et al., 2025). In our PA study, higher AIPI scores were associated with improved informational quality and expert satisfaction in multidisciplinary decision-making.

Using the AIPI, our study demonstrated a statistically significant difference in clinical reasoning depth among large language models (*p* = 0.038), with Claude Opus 4.1 (16.9 ± 2.18) and Gemini 2.5 Flash (16.4 ± 1.17) outperforming ChatGPT-5.0 (14.9 ± 1.97). A comparable reasoning–action gap was reported by Sales et al., where GPT-4 achieved high diagnostic accuracy for vestibular schwannoma (97.1%) but substantially lower accuracy in treatment recommendations (57.1%), underscoring limitations in translating correct identification into appropriate management decisions. In contrast, our findings suggest that newer Claude and Gemini models demonstrate improved alignment between diagnostic interpretation and management reasoning, indicating a narrowing of this gap in complex sellar pathology contexts (Sales et al., 2025b).

Expert satisfaction aligned with objective performance, with Claude Opus 4.1 achieving the highest AISPE-Q score (8.7 ± 0.95) and strong correlations observed between QAMAI, AIPI (*r* = 0.71), and AISPE-Q (*r* = 0.64), confirming convergent validity. Similarly, Choi et al. reported high clinician satisfaction with an AI-based clinical decision support system, with a mean satisfaction score of 8.42 ± 1.23, particularly in diagnostic reliability and clinical usefulness. Together, these findings support convergent validity between informational quality, reasoning performance, and expert trust across distinct clinical AI applications. (Choi et al., 2022). Although several between-model comparisons reached statistical significance, absolute score differences were modest and derived from ordinal rating scales, so findings should be interpreted as comparative performance signals rather than definitive evidence of clinically meaningful superiority. Clinically, these modest ordinal differences are most likely to matter through fewer omissions and clearer structuring (e.g., completeness/clarity of multidisciplinary workup and counseling language) rather than implying that any model should independently drive definitive surgical decision-making.

The assessment of LLM reliability in PA management reveals significant stratification by question complexity. Chen et al. utilized GPT-4 to evaluate 256 PA-related queries, reporting an overall reliability rate of 82.4% and a consistency rate of 68.4%. Their findings highlighted a sharp decline in performance when transitioning from general patient-oriented questions (95.0% reliability) to professional clinical queries (78.6%), with surgical management questions yielding the highest rate of completely incorrect responses (19.4%) (Chen et al., 2025). Using next-generation models and AIPI-based reasoning assessment, our study demonstrates that Claude Opus 4.1 achieves higher accuracy (4.41 ± 0.70) and completeness (4.41 ± 0.64) than previously reported GPT-4 performance, with newer models showing greater reasoning stability in complex pituitary scenarios compared with the moderate consistency described by Chen et al. (2025).

In a retrospective study of Sales et al. of 35 patients with vestibular schwannoma, GPT-4 outperformed Gemini and Bing in both diagnosis and treatment guidance. GPT-4 reached a diagnostic accuracy of 97.14%, compared to 88.57% for Gemini and 85.71% for Bing. However, the study highlights that LLMs currently remain less precise than

interdisciplinary tumor boards for definitive management planning (Sales et al., 2025b). In comparison, our study showed that higher AIPI scores were associated with more structured and consistent treatment recommendations following MRI interpretation, without implying autonomous decision-making capability.

Evaluating LLMs using real-world clinical conversations for non-critical triage BY Lee et al., Gemini 2.5 Flash achieved the highest accuracy (73.8%) and specificity (88.9%). GPT-4.1 also demonstrated balanced performance with high sensitivity (81.3%) and practical response times (1.79s). This study emphasizes that model performance varies widely between versions and suggests that few-shot prompting—providing the model with a few examples—can further enhance accuracy in clinical assessments (Lee et al., 2025). In contrast to non-critical triage, our study showed that higher-complexity tasks favored models with stronger reasoning and completeness, with Claude Opus 4.1 achieving higher AIPI (16.9 ± 2.18) and aggregate QAMAI scores.

AI and LLMs is transitioning from simple data retrieval to an adjunctive role in neuroendocrinology (Obuchowicz et al., 2025; Khosravi et al., 2024). There is significant synergy potential between LLMs and radiomics, where AI-driven prediction models can detect hormone secretion profiles from T2W MRI. LLMs like Claude Opus 4.1 can then synthesize these radiological and endocrine data points into a coherent multidisciplinary report (Zheng et al., 2025; Dai et al., 2021; Aliyeva et al., 2025; Ba et al., 2022). However, the role of AI remains supportive, not autonomous, as it lacks the "human-like" contextual communication required for sensitive patient interactions (Lee et al., 2025; Haim et al., 2025). Aksoy et al., who found Microsoft Copilot superior for general emergency questions (7.8% error margin), our results for the specific PA domain favor Claude Opus 4.1 (Aksoy and Arslan, 2025). In the study by Trecca et al., LLMs achieved a mean total QAMAI score of 22.9 ± 3.1 , performing comparably to specialists in theoretical and standardized clinical scenarios, but showed reduced performance in patient-centered questions and source transparency, underscoring that high informational quality does not uniformly translate into clinical depth or contextual trust (Vaira et al., 2024a; Trecca et al., 2025). Across models, QAMAI, AIPI, and AISPE-Q were consistently correlated, supporting convergent validity between informational quality, reasoning depth, and expert-perceived utility.

Limitations. This study is limited by the use of hypothetical scenarios and a single evaluation window, representing a time-specific snapshot of rapidly evolving LLM performance, and by reliance on text-based prompts and representative MRI images, as current LLMs lack image interpretation capabilities and remain inferior to human experts in single-image radiological assessment. Accordingly, the 'MRI-based' items should be interpreted as reasoning from limited representative images and a standardized task, not as a substitute for longitudinal radiology review or real tumor-board context. Use of simulated cases may not fully capture the variability of real clinical documentation, where inputs can be incomplete, noisy, or internally contradictory. LLM performance in practice may therefore differ when confronted with real-world chart data, nuanced comorbidities, or ambiguous imaging narratives not represented in standardized scenarios. Because this evaluation is anchored to pituitary adenoma workflows, extrapolation to other skull base or neurosurgical conditions with different imaging patterns, timelines, and decision pathways should be made cautiously.

In addition, deterministic decoding and seed control were not available through the evaluated platforms; therefore, our analysis reflects single-pass outputs and may not fully capture within-model response variability across repeated runs. Because commercial LLM providers may update models without granular public version control, the reported performance should be interpreted as a time-dependent snapshot rather than a permanent ranking. Consequently, future evaluations using the same prompt set may yield different absolute scores and potentially different between-model differences as models evolve.

Our evaluator panel consisted exclusively of board-certified skull

base specialists; ratings may preferentially reflect guideline-aligned and academically styled responses, and perceptions of usefulness or satisfaction could differ in community practice settings or among trainees.

Strengths of the Study. This study represents the first prospective, structured evaluation of pituitary adenoma management, using multiple advanced LLMs and assessed by a multidisciplinary expert panel, with three validated instruments capturing informational quality, clinical reasoning, and evaluator satisfaction. The inclusion of MRI-based scenarios and complex hormonal panels allows simultaneous assessment of diagnostic interpretation, management planning, and communication coherence within a single neuroendocrine framework.

Future Directions. Future research should focus on external validation in real-world clinical workflows, including prospective testing in multidisciplinary tumor board settings and integration of multimodal inputs such as direct MRI image analysis. The development of hybrid systems that combine Machine Learning and LLM outputs with guideline-based decision frameworks may further enhance clinical safety and reproducibility (Wang et al., 2025; Al et al., 2025; Schonfeld et al., 2024; Ong et al., 2025).

Consistent with core clinical AI governance principles, LLM use in pituitary care should remain strictly adjunctive with clearly assigned clinician responsibility for decisions, transparent documentation of AI inputs/outputs and limitations (explainability at the workflow level), and local governance controls such as access policies, audit trails, and periodic re-validation before any routine clinical integration.

Final Perspective from the Hypothesis. Consistent with the study hypothesis, significant differences were identified among LLMs in diagnostic accuracy, reasoning depth, and expert-perceived utility for managing pituitary adenomas. The consistent correlations between objective performance metrics and evaluator satisfaction support a structured, multidimensional approach to evaluating AI and LLM tools before clinical integration. Near-term pragmatic applications include guideline-anchored pre-tumor board checklists (endocrine/ophthalmologic workup prompts) and structured draft summaries for multidisciplinary discussion, with final decisions retained by clinicians.

5. Conclusion

This study demonstrates that advanced LLMs differ significantly in information quality, clinical reasoning, and evaluator satisfaction when used to manage pituitary adenomas. Among the evaluated models, Claude Opus 4.1 showed the most consistent and comprehensive performance across validated assessment tools. At the same time, Gemini 2.5 Flash and ChatGPT-5 demonstrated intermediate performance, albeit lower than that of selected domains. These findings highlight the importance of structured, multidimensional evaluation when integrating LLM and AI tools into multidisciplinary neuroendocrine care. LLMs may serve as supportive adjuncts to clinical reasoning and patient education, but their use should remain under expert oversight.

6. Integrated performance summary

Across informational accuracy, completeness, clinical reasoning, and evaluator satisfaction, Claude Opus 4.1 consistently demonstrated superior performance, characterized by higher mean scores, lower variability, stronger effect sizes, and the highest inter-rater reliability.

Gemini 2.5 Flash showed intermediate performance, with strengths in accuracy and clarity but reduced completeness and satisfaction scores. ChatGPT-5 demonstrated acceptable accuracy and clarity but lower performance in completeness and clinical reasoning domains, resulting in lower aggregate scores.

Taken together, the integrated quantitative and reliability analyses indicate that Claude Opus 4.1 provides the most robust and consistent performance across informational, clinical, and evaluative dimensions within this framework.

Informed consent

Not Applicable.

Consent to participate

Not Applicable.Consent for publication: Not Applicable.

Human participants and/or animals

None.

Authors' contributions

Conception, AA, EN, FG, MR, SM, RN.
Design, AA, EN, RN.
Supervision, EN,RN,AA.
Fundings: None;
Materials, Data, AA, EN, FG, MR, SM, RN.
Collection and/or Processing, AA, EN, FG, MR, SM, RN.
Analysis and/or Interpretation, AA, EN, FG, MR, SM, RN.
Literature, Review, AA,EN, RN.
Writing, Critical Review, AA, EN, FG, MR, SM, RN.

Funding

None.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge “Radiopaedia” for granting permission to use representative pituitary adenoma cases for educational and research purposes. We also thank the multidisciplinary evaluators for their expert contributions to case development and for rigorous participation in the evaluation process (Dr. Vafa Safarova, Dr. Vusal Najafaliyev, Antiga Muradova, and Dr. Samir Hasanov). Large language models were used only to assist with grammatical correction and structuring of the manuscript; no data analysis, interpretation, or scientific decision-making was performed by these tools.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bas.2026.105997>.

Data availability

De-identified study materials are available from the corresponding author upon reasonable request, including the anonymized LLM outputs, per-question domain scores, and individual evaluator ratings for QAMAI, AIPI, and AISPE-Q.

References

- Akinci D'Antonoli, T., Stanzione, A., Bluethgen, C., et al., 2024. Large language models in radiology: fundamentals, applications, ethical considerations, risks, and future directions. *Diagn. Interv. Radiol.* 30 (2), 80–90. <https://doi.org/10.4274/dir.2023.232417>.
- Aksoy, I., Arslan, M.K., 2025. Comparison of performance of artificial intelligence tools in answering emergency medicine question pool: ChatGPT 4.0, google gemini and

- microsoft copilot. *Pakistan J. Med. Sci.* 41 (4), 968–972. <https://doi.org/10.12669/pjms.41.4.11178>.
- Al, Zaabi A., Alshibli, R., AlAmri, A., AlRuheili, I., Lutfi, S.L., 2025. Trends and trajectories in the rise of large language models in radiology: scoping review. *JMIR Med. Inform.* 13, e78041. <https://doi.org/10.2196/78041>. Published 2025 Dec 9.
- Alharbi, T.A.F., Rababa, M., Alsuwayl, H., Alsubail, A., Alenizi, W.S., 2025. Diagnostic challenges and patient safety: the critical role of accuracy - a systematic review. *J. Multidiscip. Healthc.* 18, 3051–3064. <https://doi.org/10.2147/JMDH.S512254>. Published 2025 May 30.
- Aliyeva, A., Alaskarov, E., Sari, E., 2025. Postoperative management of tympanoplasty with ChatGPT-4.0. *J Int Adv Otol* 21 (1), 1–6. <https://doi.org/10.5152/iao.2025.241797>.
- Aliyeva, A., Muradova, A., Hashimli, R., Müderris, T., 2026. Multi-model artificial intelligence evaluation in sudden sensorineural hearing loss. *Otolaryngol. Head Neck Surg.* 28. <https://doi.org/10.1002/ohn.70143>. Published online January.
- Aluyi-Osa, G., Suleman, A., Salati, C., et al., 2025. Multidisciplinary management of pituitary macroadenoma. *World J. Methodol.* 15 (3), 97694. <https://doi.org/10.5662/wjmv15.i3.97694>. Published 2025 Sep. 20.
- Anthropic, 2025. Claude opus (version 4.5) [large language model]. <https://claude.ai/>.
- Baysal, B., Eser, M.B., Dogan, M.B., Kursun, M.A., 2022. Multivariable diagnostic prediction model to detect hormone secretion profile from T2W MRI radiomics with artificial neural networks in pituitary adenomas. *Medeni Med J* 37 (1), 36–43. <https://doi.org/10.4274/MMJ.galenos.2022.58538>.
- Banskota, S., Adamson, D.C., 2021. Pituitary adenomas: from diagnosis to therapeutics. *Biomedicines* 9 (5), 494. <https://doi.org/10.3390/biomedicines9050494>. Published 2021 Apr 30.
- Chen, Y., Chen, Y., Chen, L., Feng, T., Wang, S., 2025. Evaluation of ChatGPT as a supplementary tool for pituitary adenomas: an observational study based on simulated consultations. *Medicine (Baltim.)* 104 (46), e45928. <https://doi.org/10.1097/MD.00000000000045928>.
- Choi, I.H., Seo, G.H., Park, J., et al., 2022. Evaluation of users' level of satisfaction for an artificial intelligence-based diagnostic program in pediatric rare genetic diseases. *Medicine (Baltim.)* 101 (28), e29424. <https://doi.org/10.1097/MD.00000000000029424>. Published 2022 Jul 15.
- Dai, C., Sun, B., Wang, R., Kang, J., 2021. The application of artificial intelligence and machine learning in pituitary adenomas. *Front. Oncol.* 11, 784819. <https://doi.org/10.3389/fonc.2021.784819>. Published 2021 Dec 23.
- Fahim, Y.A., Hasani, I.W., Kabba, S., Ragab, W.M., 2025. Artificial intelligence in healthcare and medicine: clinical applications, therapeutic advances, and future perspectives. *Eur. J. Med. Res.* 30 (1), 848. <https://doi.org/10.1186/s40001-025-03196-w>. Published 2025 Sep. 23.
- Gaillard F, Pituitary adenoma - involving clivus. Case study, radiopaedia.org (Accessed on 22 December 2025).<https://doi.org/10.53347/rId-5578https://radiopaedia.org/cases/pituitary-adenoma-involving-clivus?lang=us>.
- Google, DeepMind, 2025. Gemini 2.5 flash (version 2.5 flash) [large language model]. <https://gemini.google.com/>.
- Haim, G.B., Saban, M., Barash, Y., Cirulnik, D., Shaham, A., Eisenman, B.Z., Burshtein, L., Mymon, O., Klang, E., 2025. Evaluating large language model-assisted emergency triage: a comparison of acuity assessments by GPT-4 and medical experts. *J. Clin. Nurs.* <https://doi.org/10.1111/jocn.17490>.
- Jesser, J., Schlamp, K., Bendszus, M., 2014. Hypophysentumoren [pituitary gland tumors]. *Radiologe* 54 (10), 981–988. <https://doi.org/10.1007/s00117-014-2688-5>.
- Karimian-Jazi, K., 2019. Hypophysentumoren [pituitary gland tumors]. *Radiologe* 59 (11), 982–991. <https://doi.org/10.1007/s00117-019-0570-1>.
- Khosravi, M., Zare, Z., Mojtabaiean, S.M., Izadi, R., 2024. Artificial intelligence and decision-making in healthcare: a thematic analysis of a systematic review of reviews. *Health Serv Res Manag Epidemiol* 11, 23333928241234863. <https://doi.org/10.1177/23333928241234863>. Published 2024 Mar 5.
- Lechien, J.R., Maniaci, A., Gengler, I., Hans, S., Chiesa-Estomba, C.M., Vaira, L.A., 2024. Validity and reliability of an instrument evaluating the performance of intelligent chatbot: the artificial intelligence performance instrument (AIPI). *Eur. Arch. Otorhinolaryngol.* 281 (4), 2063–2079. <https://doi.org/10.1007/s00405-023-08219-y>.
- Lee, S., Jung, S., Park, J.H., et al., 2025. Performance of ChatGPT, gemini and DeepSeek for non-critical triage support using real-world conversations in emergency department. *BMC Emerg. Med.* 25, 176. <https://doi.org/10.1186/s12873-025-01337-2>.
- Lin, A., Wang, Z., Jiang, A., et al., 2025. Large language models in clinical trials: applications, technical advances, and future directions. *BMC Med.* 23 (1), 563. <https://doi.org/10.1186/s12916-025-04348-9>. Published 2025 Oct 14.
- Maity, S., Saikia, M.J., 2025. Large language models in healthcare and medical applications: a review. *Bioengineering (Basel)* 12 (6), 631. <https://doi.org/10.3390/bioengineering12060631>. Published 2025 Jun 10.
- Merchant, S.A., Merchant, N., Varghese, S.L., Shaikh, M.J.S., 2025. Large language models and large concept models in radiology: present challenges, future directions, and critical perspectives. *World J. Radiol.* 17 (11), 114754. <https://doi.org/10.4329/wjr.v17.i11.114754>.
- Neuropathology R, Invasive gonadotroph cell pituitary adenoma. Case study, Radiopaedia.org (Accessed on 22 December 2025).<https://radiopaedia.org/cases/invasive-gonadotroph-cell-pituitary-adenoma?lang=us> title="https://doi.org/10.53347/rId-29634 https://radiopaedia.org/cases/invasive-gonadotroph-cell-pituitary-adenoma?lang=us"><https://doi.org/10.53347/rId-29634> https://radiopaedia.org/cases/invasive-gonadotroph-cell-pituitary-adenoma?lang=us.
- O'Donnell C, Cystic pituitary adenoma. Case study, radiopaedia.org (Accessed on 22 December 2025).<https://doi.org/10.53347/rId-18097> https://radiopaedia.org/cases/cystic-pituitary-adenoma?lang=us.

- Obuchowicz, R., Lasek, J., Wodziński, M., Piórkowski, A., Strzelecki, M., Nurzynska, K., 2025. Artificial intelligence-empowered radiology-current status and critical review. *Diagnostics (Basel)* 15 (3), 282. <https://doi.org/10.3390/diagnostics15030282>. Published 2025 Jan 24.
- Ong, J.C.L., Jin, L., Elangovan, K., et al., 2025. Large language model as clinical decision support system augments medication safety in 16 clinical specialties. *Cell Rep. Med.* 6 (10), 102323. <https://doi.org/10.1016/j.xcrm.2025.102323>.
- OpenAI, 2025. ChatGPT (GPT-5 version) [large language model]. <https://chat.openai.com/>.
- Patil, A., Serrato, P., Chisvo, N., Arnaout, O., See, P.A., Huang, K.T., 2024. Large language models in neurosurgery: a systematic review and meta-analysis. *Acta Neurochir (Wien)* 166 (1), 475. <https://doi.org/10.1007/s00701-024-06372-9>. Published 2024 Nov 23.
- Pourhoseingholi, M.A., Killan, C., Rafiee, S., Hoare, D.J., Wray, N., Bateman, P., 2025. Validity, reliability, and readability of artificial intelligence chatbots as public sources of information on hearing loss: a comparative evaluation of ChatGPT, Bing, Gemini, and Perplexity. *Int. J. Audiol.* <https://doi.org/10.1080/14992027.2025.2569927>. Published online.
- Rasuli B, Pituitary macroadenoma. Case study, radiopaedia.org (Accessed on 22 December 2025). <https://doi.org/10.5334/rID-67287>. <https://radiopaedia.org/cases/pituitary-macroadenoma-32?lang=us>.
- Rosen, A.W., Ose, I., Gögenur, M., et al., 2025. Clinical implementation of an AI-based prediction model for decision support for patients undergoing colorectal cancer surgery. *Nat. Med.* 31, 3737–3748. <https://doi.org/10.1038/s41591-025-03942-x>.
- Sales, A.H.A., Gizaw, C.J., Beck, J., Grauvogel, J., 2025a. Evaluating large language models in interpreting MRI reports and recommending treatment for vestibular schwannoma. *Diagnostics (Basel)* 15 (22), 2841. <https://doi.org/10.3390/diagnostics15222841>. Published 2025 Nov 10.
- Sales, A.H.A., Gizaw, C.J., Beck, J., Grauvogel, J., 2025b. Evaluating large language models in interpreting MRI reports and recommending treatment for vestibular schwannoma. *Diagnostics (Basel)* 15 (22), 2841. <https://doi.org/10.3390/diagnostics15222841>. Published 2025 Nov 10.
- Savage, S.A., Seth, I., Angus, Z.G., Rozen, W.M., 2025. Advancements in microsurgery: a comprehensive systematic review of artificial intelligence applications. *J. Plast. Reconstr. Aesthetic Surg.* 101, 65–76.
- Schonfeld, E., Mordekai, N., Berg, A., et al., 2024. Machine learning in neurosurgery: toward complex inputs, actionable predictions, and generalizable translations. *Cureus* 16 (1), e51963. <https://doi.org/10.7759/cureus.51963>. Published 2024 Jan 9.
- Trecca, E.M.C., Caponio, V.C.A., Turri-Zanoni, M., di Lullo, A.M., Gaffuri, M., Lechien, J. R., Maniaci, A., Maruccio, G., Reale, M., Visconti, I.C., Dallari, V., 2025. Comparative analysis of information quality in pediatric otorhinolaryngology: clinicians, residents, and large language models. *Otolaryngol. Head Neck Surg.* 173, 228–236. <https://doi.org/10.1002/ohn.1225>.
- Tsoriev A, Rathke's cleft cyst and ACTH secreting pituitary microadenoma: simultaneous symptomatic. Case study, Radiopaedia.org. (Accessed on 22 December 2025) <https://doi.org/10.5334/rID-34272>. <https://radiopaedia.org/cases/rathkes-cleft-cyst-and-acth-secreting-pituitary-microadenoma-simultaneous-symptomatic?lang=us>.
- Vaira, L.A., Lechien, J.R., Abbate, V., et al., 2024a. Validation of the quality analysis of medical artificial intelligence (QAMAI) tool: a new tool to assess the quality of health information provided by AI platforms. *Eur. Arch. Otorhinolaryngol.* 281 (11), 6123–6131. <https://doi.org/10.1007/s00405-024-08710-0>.
- Vaira, L.A., Lechien, J.R., Abbate, V., et al., 2024b. Accuracy of ChatGPT-Generated information on head and neck and oromaxillofacial surgery: a multicenter collaborative analysis. *Otolaryngol. Head Neck Surg.* 170 (6), 1492–1503. <https://doi.org/10.1002/ohn.489>.
- Wang, J., Shue, K., Liu, L., et al., 2025. Preliminary evaluation of ChatGPT model iterations in emergency department diagnostics. *Sci. Rep.* 15, 10426. <https://doi.org/10.1038/s41598-025-95233-1>.
- Zheng, A., Tang, D., He, H., Liang, X., 2025. Artificial intelligence-driven approaches in pituitary neuroendocrine tumors: integrating endocrine-metabolic profiling for enhanced diagnostics and therapeutics. *Front Endocrinol (Lausanne)* 16, 1618412. <https://doi.org/10.3389/fendo.2025.1618412>. Published 2025 Oct 16.